

1 October 2026

The Editor-in-Chief, *Neurocomputing*

Dear Editor,

I am submitting “**FOLoRA: Fisher-Orthogonalized Low-Rank Adaptation for Continual Learning**” for consideration as a research paper. It is original work, not under consideration elsewhere, and has a single author.

The gap. Continual learning with pre-trained models is now led by parameter-efficient methods. Within them, one line — orthogonalization and subspace allocation — decides *where* a new task may write the weights: O-LoRA orthogonally initializes each task’s adapter against earlier tasks’ input subspaces, and InfLoRA places the new subspace in the intersection of its gradient space with the orthogonal complement of the old tasks’. Both inherit two choices from the classical gradient-projection literature: the protected subspace comes from a *structural* criterion (an input covariance, a gradient subspace, an explicit orthogonality constraint) rather than from the curvature of the past loss, and once chosen, every retained direction is defended *equally* — the budget decides how many directions to keep, never how much each one matters. This manuscript starts from the observation that the second choice is not forced by anything.

What the paper does. I derive a second-order forgetting bound for sequential low-rank adaptation, with an explicit higher-order remainder, and show that its leading term is a Fisher-weighted projection of the adapter displacement onto the past tasks’ eigendirections. That term is exactly what a rank- k regularizer can minimize, which yields the proposed objective: keep one fixed-budget adapter, estimate a per-layer importance kernel from the empirical Fisher restricted to it, and penalize displacement along its top eigendirections in proportion to their curvature. The equal-weight geometry underlying O-LoRA-style methods is recovered as the special case of equal eigenvalues, so the contribution generalizes the existing subspace family rather than competing with it. The method is replay-free, adds no inference cost (the update merges back into the frozen backbone), and keeps the trainable parameter count constant in the number of tasks, unlike per-task-adapter designs.

Why it is not incremental, against the closest work. The contribution is deliberately narrow and therefore checkable. It is not an accuracy claim against EWC: the paper reports *parity* with the strongest regularized baseline and says so. The positive results are (i) against the parameter-efficient family it belongs to — L2P, CODA-Prompt, InfLoRA, O-LoRA — where paired differences are significant on both benchmarks and of one sign across seeds; (ii) an ablation isolating the mechanism: removing the Fisher-weighted penalty and changing nothing else returns the method to unprotected sequential fine-tuning, so the entire effect is attributable to the term the derivation identifies; and (iii) evidence that the protected rank is a memory–accuracy dial rather than a sensitive hyperparameter. The theory is offered as a statement of *which* quantity to penalize, not as a numerical predictor — the manuscript measures the bound’s slack and reports that it is not tight.

Scope. The work is a learning rule: a training objective derived for a parameter-efficient continual learner and evaluated on standard class-incremental benchmarks under the frozen-feature nearest-class-mean protocol used by the methods it compares against. That places it in the journal’s interest in learning algorithms and training objectives, and it engages explicitly with recent *Neurocomputing* work on task-adaptive and subspace-based continual learning, gradient-based task organization, and evaluation-protocol artifacts in class-incremental learning, all of which are cited and discussed in the manuscript.

Two places where the paper reports against its own interest — the parts I would most want a referee to check. First, it reports the reproducibility floor it measured (1.29 accuracy points at a fixed seed and configuration) and then applies it. The difference against the strongest regularized baseline falls *inside* that band on the primary benchmark, so that comparison is reported as parity rather than a win, and the baseline’s full regularization curve is printed because the sign of the comparison flips across it — including the operating points at which that baseline has the *smaller* CIL–TIL gap than the proposed method. Second, it reports that on one benchmark two reproduced baselines land at or below the frozen-feature floor, with the mechanism the author believes is responsible, rather than reporting only the variants that flatter the comparison. Where a baseline was run on fewer seeds than the proposed method, the table states each budget rather than implying a shared one. The positive accuracy comparisons are the ones against the parameter-efficient family, and they are

stated as such, not as a win against full-parameter regularization. Code, protocol and the scripts that regenerate every table and figure are openly available; the datasets are public.

Suggested reviewers. These researchers have recently published closely related work in *Neurocomputing*, which is why I suggest them; I would welcome any alternative the editorial office prefers.

- Hongwei Zhao — task-adaptive LoRA teachers for continual learning.
- Lei Pan — task-adaptive continual learning of vision–language models via prototype routing and prompts.
- Weizhuo Zhang — class-incremental learning under classifier bias and calibration.
- Junlong Huang — reducing catastrophic forgetting via gradient-based task organization.
- Xulong Wang — continual learning for dissimilar tasks via gradient-norm regularization.

Declarations. No competing financial interests or personal relationships could have influenced this work; no external funding; datasets publicly available; code and reproduction scripts openly available.

Thank you for your time and consideration.

Sincerely,
Jiang Tao